

APOLLO INSTITUTE OF COMPUTER APPLICATION

BACHELOR OF COMPUTER APPLICATION

SEMESTER 03

Design Thinking and Innovation (BC03001071)

Assignment 1

Unit 1: Introduction to Design Thinking

1) What is Design Thinking? Explain its attributes.

Introduction

Attributes of Design Thinking are the key characteristics or qualities that make Design Thinking an effective approach to solving problems. These attributes help designers and organizations create innovative, practical, and user-friendly solutions. They ensure that the final product or service not only solves the problem but also meets the needs and expectations of users.

1. Human-Centered

Meaning:

Human-Centered Design means that the entire design process focuses on the people who will use the product or service. Instead of starting with technology or business goals, designers first understand the users' needs, behaviors, expectations, and challenges.

The objective is to create solutions that are useful, easy to use, and provide a positive user experience.

Key Points

- Focuses on users rather than technology.
- Identifies real user problems.
- Improves customer satisfaction.
- Makes products more accessible and user-friendly.

Example

While designing a **mobile banking app for elderly users**, developers use large buttons, simple menus, high-contrast colors, and voice guidance so that senior citizens can use the app

comfortably.

2. Empathy

Meaning:

Empathy is the ability to understand and share the feelings, experiences, and perspectives of users. It is one of the most important attributes of Design Thinking because it helps designers identify problems from the users' point of view.

Designers gather information through interviews, observations, surveys, and discussions.

Key Points

- Understands users' emotions and needs.
- Helps identify hidden problems.
- Builds trust with users.
- Leads to better and more relevant solutions.

Example

Before developing an **online learning platform**, designers interview students and teachers to understand issues such as poor internet connectivity, confusing interfaces, and difficulty accessing study materials.

3. Creativity

Meaning:

Creativity is the ability to generate new, original, and useful ideas. Design Thinking encourages teams to think beyond traditional solutions and explore multiple possibilities before selecting the best one.

There is no restriction on ideas during brainstorming; even unusual ideas are welcomed.

Key Points

- Encourages innovative thinking.
- Generates multiple possible solutions.
- Avoids limiting ideas too early.
- Promotes continuous innovation.

Example

A bicycle manufacturer designs a **foldable bicycle** for people living in crowded cities, making it easier to carry and store in small apartments or public transport.

4. Collaboration

Meaning:

Collaboration means working together with people from different backgrounds, skills, and expertise. Since every person has a different perspective, teamwork often results in more creative and practical solutions.

Design Thinking encourages open communication and sharing of ideas.

Key Points

- Promotes teamwork.
- Combines diverse knowledge and skills.
- Improves decision-making.
- Produces better solutions.

Example

To develop a **healthcare application**, doctors provide medical knowledge, software developers build the application, designers create an easy-to-use interface, and patients provide feedback about usability.

5. Experimentation

Meaning:

Experimentation involves trying different ideas by creating simple prototypes and testing them before developing the final product. This helps identify problems early and reduces the risk of failure.

Experimentation encourages learning through practical experience.

Key Points

- Tests different ideas.
- Identifies mistakes early.
- Reduces development cost.
- Improves product quality.

Example

A company creates **three different website layouts** and asks users to test them. Based on user preferences and feedback, the best design is selected.

6. Iteration

Meaning:

Iteration means continuously improving a solution by testing it, collecting feedback, and making necessary changes. Design Thinking does not assume that the first solution is perfect. Instead, products are refined until they meet user expectations.

Design Cycle

Prototype

↓

Test

↓

Collect Feedback

↓

Improve

↓

Retest

Key Points

- Continuous improvement.
- Uses user feedback.
- Corrects errors quickly.
- Produces high-quality solutions.

Example

A mobile application releases Version 1.0, collects customer reviews, fixes problems, adds new features, and launches Version 2.0.

7. Optimism

Meaning:

Optimism is the belief that every problem has a solution. Design Thinking encourages teams to remain positive even when facing difficult challenges or unsuccessful attempts.

A positive mindset motivates teams to continue experimenting until they discover an effective solution.

Key Points

- Encourages confidence.
- Motivates continuous improvement.
- Supports innovation.
- Accepts challenges positively.

Example

A startup developing a water purification system continues improving its prototype even after several failed tests because the team believes they can eventually create an affordable solution.

8. User Feedback

Meaning:

User feedback is the information and suggestions collected from users after they interact with a product or prototype. Feedback helps designers understand what works well and what needs improvement.

Continuous feedback ensures that the final product truly meets user expectations.

Key Points

- Helps improve products.
- Identifies user satisfaction.
- Reduces design errors.
- Encourages continuous improvement.

Example

A food delivery app introduces a new interface and asks customers to rate their experience. Based on the feedback, the navigation and payment process are simplified.

9. Practicality

Meaning:

Practicality means that a solution should be realistic, affordable, and easy to implement. An innovative idea is valuable only if it can be developed within available resources and effectively solve the intended problem.

Key Points

- Solutions should be feasible.
- Considers budget and available technology.

- Easy to implement and maintain.
- Solves real-world problems.

Example

Instead of building an expensive smart irrigation system, engineers develop a **low-cost automatic watering system** that small farmers can easily afford and maintain.

10. Innovation

Meaning:

Innovation means creating new or improved products, services, or processes that provide value to users. Innovation is not always about inventing something completely new; it can also involve improving existing solutions.

Key Points

- Creates value for users.
- Improves existing products.
- Encourages competitive advantage.
- Solves problems in better ways.

Example

The introduction of **UPI (Unified Payments Interface)** transformed digital payments in India by making money transfers instant, secure, and convenient using smartphones.

2) Explain models of Design Thinking in detail.

Introduction

A **Design Thinking Model** is a structured framework that guides designers and innovators through the process of understanding users, identifying problems, generating ideas, building solutions, and improving them through testing.

Different organizations have developed different models of Design Thinking. Although the number of stages may vary, all models focus on **human-centered problem solving, creativity, innovation, and continuous improvement.**

The four most popular Design Thinking models are:

1. Stanford d.school Model
 2. IDEO Model
 3. Double Diamond Model
 4. IBM Design Thinking Model
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1. Stanford d.school Model

The **Stanford d.school Model** is the most widely used Design Thinking model. It consists of **five stages**.

Diagram

Empathize



Define



Ideate



Prototype



Test

Stage 1: Empathize

- Understand the users and their needs.
- Observe, interview, and interact with users.
- Gather information about their problems, emotions, and experiences.

Example:

Interview students to understand why they struggle with online learning.

Stage 2: Define

- Analyze the collected information.
- Clearly identify the actual problem.
- Prepare a problem statement.

Example:

"Students cannot easily access study materials anytime."

Stage 3: Ideate

- Generate many possible solutions.
- Use brainstorming and creative thinking.
- Do not reject ideas during this stage.

Example:

Develop a mobile learning app, digital library, or AI tutor.

Stage 4: Prototype

- Create a simple model of the selected solution.
- It can be a paper sketch, wireframe, mock-up, or working model.

Example:

Design the first version of a mobile learning application.

Stage 5: Test

- Give the prototype to users.
- Collect feedback.
- Improve the solution if required.

Example:

Students test the app and suggest adding offline access.

2. IDEO Design Thinking Model

The **IDEO Model** was developed by the global design company **IDEO**. It focuses on understanding users, encouraging innovation, and implementing practical solutions.

Diagram

Inspiration



Ideation



Implementation

Phase 1: Inspiration

- Understand users.
- Observe real-life situations.
- Identify opportunities and problems.

Example:

Observe patients in hospitals to understand their difficulties.

Phase 2: Ideation

- Generate different ideas.
- Select the best solution.
- Create prototypes.

Example:

Design an appointment booking application.

Phase 3: Implementation

- Develop the final product.
- Launch it.
- Continue improving using customer feedback.

Example:

Release the hospital app and update it regularly.

3. Double Diamond Model

The **Double Diamond Model** was developed by the **UK Design Council**. It consists of **four stages** and emphasizes both **divergent thinking** (exploring many possibilities) and **convergent thinking** (selecting the best option).

Diagram

Discover → Define → Develop → Deliver

Stage 1: Discover

- Research users.
- Collect information.
- Understand the problem.

Example:

Interview customers about difficulties while shopping online.

Stage 2: Define

- Analyze research findings.
- Identify the exact problem.

Example:

Customers find the checkout process too complicated.

Stage 3: Develop

- Generate different ideas.
- Create prototypes.
- Experiment with possible solutions.

Example:

Design a one-click checkout option.

Stage 4: Deliver

- Test the solution.
- Launch the final product.
- Monitor user feedback and improve.

Example:

Release the improved shopping app.

4. IBM Design Thinking Model

The **IBM Design Thinking Model** combines user-centered design with teamwork and continuous improvement. It is widely used in software and IT industries.

IBM focuses on three important principles:

- **Hills**
- **Playbacks**
- **Sponsor Users**

1. Hills

- Clearly define user goals.
- Focus on solving users' real problems.

Example:

"Our app should allow users to book tickets within two minutes."

2. Playbacks

- Regularly review progress with team members and users.
- Share ideas and receive feedback.

Example:

Developers demonstrate each software version to users.

3. Sponsor Users

- Real users participate throughout the project.
- Their feedback helps improve the product.

Example:

College students continuously test an educational application during development.

3) Explain in detail the four main principles of Design Thinking. Support your answer with suitable examples.

Introduction

Design Thinking is a human-centered approach to solving problems that focuses on understanding users, generating innovative ideas, and developing effective solutions. It follows certain guiding principles that help designers create products and services that are useful, practical, and user-friendly.

The **four main principles of Design Thinking** are:

1. Human-Centeredness
2. Collaboration
3. Experimentation
4. Iteration

These principles encourage creativity, teamwork, continuous learning, and innovation.

1. Human-Centeredness (User-Centered Approach)

Definition

Human-Centeredness means **placing the user at the center of the design process**. Instead of making assumptions, designers study users' needs, emotions, behaviors, and challenges to create solutions that truly solve their problems.

Key Features

- Focuses on users rather than technology.
- Uses empathy to understand users.
- Identifies real problems through observation and interviews.
- Improves user satisfaction.

Example

A college develops a **student mobile application**.

Instead of simply adding many features, the development team first interviews students and finds that they mainly need:

- Easy attendance tracking
- Exam timetable
- Assignment reminders

The final app focuses on these important features, making it more useful and user-friendly.

Benefits

- Better customer satisfaction
 - Easy-to-use products
 - Higher acceptance of solutions
-

2. Collaboration

Definition

Collaboration means **working together with people from different backgrounds and areas of expertise** to generate better ideas and solutions.

A Design Thinking team may include:

- Designers
- Software developers
- Business experts
- Marketing professionals
- Customers

Each member contributes a unique perspective.

Key Features

- Encourages teamwork.

- Combines technical and creative ideas.
- Improves communication.
- Leads to innovative solutions.

Example

While designing an **online shopping website**:

- Developers create the software.
- Designers improve the user interface.
- Marketing experts study customer preferences.
- Customers provide feedback during testing.

Together, they create a website that is attractive, easy to use, and meets customer needs.

Benefits

- Better decision-making
 - Increased creativity
 - Faster problem-solving
-

3. Experimentation

Definition

Experimentation means **trying different ideas, creating prototypes, testing them, and learning from the results** before developing the final product.

Instead of assuming one solution is correct, designers explore multiple possibilities.

Key Features

- Encourages creativity.
- Supports learning through testing.
- Reduces project risks.

- Helps discover better solutions.

Example

A company wants to develop a **food delivery application**.

Instead of directly launching the final app, the team creates three different homepage designs.

They ask users to test each version and identify which one is easiest to use. Based on the feedback, they select the best design.

Benefits

- Early identification of mistakes
 - Lower development cost
 - Better-quality products
-

4. Iteration

Definition

Iteration means **continuously improving a product or solution based on testing and user feedback**. Design Thinking is not a one-time process; it involves repeating the design cycle until the best solution is achieved.

Key Features

- Continuous improvement.
- Regular user feedback.
- Flexible design process.
- Better product quality.

Example

A mobile banking application is launched.

After receiving customer feedback, the developers add:

- Fingerprint login
- Dark mode
- Faster payment options
- Improved security

Each update makes the application more useful and user-friendly.

Benefits

- Continuous innovation
 - Increased customer satisfaction
 - Long-term product improvement
-

4) Explain the origin of Design Thinking.

Introduction

Design Thinking is a **human-centered approach** to solving problems through creativity, innovation, empathy, and experimentation. It focuses on understanding users' needs before developing products or services. Although the term "Design Thinking" became popular in recent decades, its roots can be traced back to the **1960s and 1970s**, when researchers began studying how designers think while solving complex problems.

Today, Design Thinking is widely used in business, engineering, healthcare, education, software development, and social innovation.

Origin of Design Thinking

The origin of Design Thinking can be understood through its historical development.

1. Beginning in the 1960s

The concept of Design Thinking began in the **1960s** when researchers tried to understand how designers solve complex problems.

Unlike traditional problem-solving, designers used:

- Creativity
- Observation
- Experimentation
- User-focused thinking

This led to the idea that design methods could also solve business and social problems.

2. Herbert A. Simon (1969)

The first major contribution came from **Herbert A. Simon**, who introduced the concept in his famous book "**The Sciences of the Artificial**" (1969).

Simon explained that:

- Design is a systematic process.
- Problems should be analyzed before creating solutions.
- Decision-making and creativity are both important in design.

He is often regarded as one of the **founders of Design Thinking** because he showed that design follows a structured process rather than relying only on artistic talent.

Contribution:

- Introduced systematic problem-solving in design.
 - Connected design with science and decision-making.
-

3. Robert McKim (1973)

In 1973, Robert McKim published the book "**Experiences in Visual Thinking.**"

He emphasized:

- Creative thinking
- Visualization
- Brainstorming
- Sketching ideas

His work encouraged designers to use visual methods for solving problems and inspired many future Design Thinking practices.

4. Peter Rowe (1987)

The term "**Design Thinking**" became more widely recognized after **Peter Rowe** published the book "**Design Thinking**" in 1987.

He studied how architects and designers:

- Analyze problems
- Generate ideas
- Make design decisions

His work formally introduced the phrase **Design Thinking** into academic literature.

5. IDEO and David Kelley (1990s)

In the **1990s**, the design consultancy **IDEO**, led by **David Kelley**, made Design Thinking popular in industries.

IDEO developed a process that focused on:

- Empathy

- Brainstorming
- Rapid prototyping
- Testing solutions

This approach helped companies create innovative products and services based on customer needs.

Example:

IDEO redesigned shopping carts and healthcare services by studying users and testing multiple prototypes before finalizing the design.

6. Stanford d.school (2005)

In **2005**, **Stanford University** established the **Hasso Plattner Institute of Design**, commonly known as the **d.school**.

The d.school developed the famous **five-stage Design Thinking model**:

Empathize

↓

Define

↓

Ideate

↓

Prototype

↓

Test

This model is now widely used in universities, businesses, and innovation centers across the world.

7. Modern Applications

Today, Design Thinking is applied in many fields, including:

- Software Development
- Mobile App Development
- Healthcare
- Education
- Banking
- Business Strategy
- Engineering
- Government Services
- Social Innovation

Companies use Design Thinking to develop products that are innovative, user-friendly, and effective.